Nagar YuwakShikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to RashtrasantTukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
NAAC A++
Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department *To nurture computer engineering professionals by offering:*

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration • Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Hrutik Gujarkar
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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15/07/2026

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Session	**2025-26 (EVEN)**	**Course Name**	**LAB: Optimization Techniques**
Semester	**7th**	**Course Code**	**23ADS1707**
Roll No	***39***	**Name of Student**	**Hrutik Gujarkar**

Practical Number	**03**
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To solve constrained optimization problems involving equality constraints using the Direct Substitution Method.
Problem Definition	Constrained Optimization Using Direct Substitution Method

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Theory (100 words)	In **constrained optimization**, an objective function is optimized while satisfying one or more constraints. Given: [\text{Optimize } f(x,y)] Subject to: [g(x,y)=0] In the **Direct Substitution Method**, one variable is expressed in terms of the other using the constraint equation and substituted into the objective function. This reduces the problem to a **single-variable unconstrained optimization problem**. The resulting function is optimized by finding critical points using the **first derivative test** and verifying them using the **second derivative test**.

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	Problem Statement Optimize: [f(x,y)] Subject to: [g(x,y)=0]

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Procedure and Execution (100 Words)	Algorithm: 1. Define the objective function and constraint. 2. Express one variable in terms of the other using the constraint equation. 3. Substitute the expression into the objective function. 4. Obtain a single-variable function. 5. Differentiate the function with respect to the remaining variable. 6. Equate the derivative to zero and find the critical point. 7. Compute the second derivative. 8. Determine whether the point is minimum or maximum. 9. Find the corresponding value of the eliminated variable. 10. Display the optimal solution.

	Code:
	
	`importsympyassp`
	
	`# Define variable`
	`x = sp.Symbol('x')`
	
	`# Constraint: x + y = 10`
	`y = 10 - x`
	
	`# Objective function`
	`f = x**2 + y**2`
	`print("Objective Function after substitution:") print(sp.expand(f))`

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```
# First derivative
f_prime = sp.diff(f, x)
 print("\nFirst
Derivative:")
print(f_prime)

# Critical point critical_point =
sp.solve(f_prime, x)

# Second derivative f_double
= sp.diff(f_prime, x)
 for point incritical_point:
y_value = y.subs(x, point)
     print("\nOptimal
Point:")      print("x =",
point)      print("y
=",y_value)
     iff_double.subs(x, point)>0:
print("Minimum Point")
eliff_double.subs(x, point)<0:
        print("Maximum Point")
else:
        print("Test Inconclusive")
     print("Objective Function Value =",f.subs(x,
point))
```

Output:

```
Objective Function after substitution:
2*x**2 - 20*x + 100
First Derivative:
4*x - 20

Optimal
Point:
x   =
5   y
= 5
Minimum
Point
```

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```
Objective Function Value = 50
```

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Output Analysis	The objective function was simplified after substitution to (2x^2-20x+100). Its first derivative, (4x-20), was set to zero, giving the optimal value (x=5). Using the constraint, the corresponding value is (y=5). Evaluating the objective function at this point gives a minimum value of **50**, confirming that the optimal solution is **((5,5))**.
Link of student Github profile where lab assignment has been uploaded	-Technique-Practicals
Conclusion	The optimal point is **((5,5))**, where the objective function attains its **minimum value of 50**.
Plag Report (Similarity index < 12%)	
Date	**15/07/2026**